Alignment Is Not Enough for Safe Medical LLM Evaluation

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Abstract

LLM-as-judges (LLJs) are increasingly used to scale medical AI evaluation and reduce clinician review costs. However, high alignment with clinicians alone does not establish evaluation validity, reliability, or safety. This position paper argues that LLJ–clinician alignment should not be reported as standalone evidence that a judge is reliable, clinically grounded, or safe for downstream use. Through multiple medical tasks, we identify four risks of existing medical LLJ evaluation: unstable evaluations across prompts, runs, and judge models; opaque scores that hide clinically meaningful errors; weak validity under clinical uncertainty; and downstream safety and data contamination risks. By re-examining LLJ evaluation setups in prior medical studies, we find that repeated identical judge calls reproduce only 66.2–83.8% of original ratings, while swapping multiple-choice answer order changes 11–40% of paired preferences. Additionally, changing the clinical diagnosis matching rule changes diagnostic accuracy by 18.8 percentage points with the same model predictions. To address these limitations, we propose a six-step medical LLJ evaluation protocol with four readiness levels that define minimum assurance requirements and use restrictions. The protocol combines clinician-verified codebook development, validation on representative subsets, and adjudication of uncertain or high-severity cases, with clinician involvement scaled to the consequences of evaluation errors to support scalable medical evaluation.

1 Introduction

Large language models (LLMs) are increasingly used not only to generate medical text [1, 2, 3, 4], but also to evaluate it [5, 6, 7]. These LLM-as-judges (LLJs) systems create a second-order safety problem: medical AI systems may be benchmarked, selected, or trusted based on judgments produced by other models. In general-domain natural language generation (NLG) tasks, an unreliable judge may distort leaderboard rankings and model development [6]. In medicine, however, the evaluation error may provide an unsafe answer to patients [8, 9], reward unsupported clinical content [3, 4, 10, 11], reject a clinically valid diagnosis [12, 13], or misrank models considered for downstream applications use [14, 15]. Recent general-domain work has raised concerns that LLJs may be neither valid nor reliable as NLG evaluators, emphasizing problems such as inconsistent judgment collection, weak construct validity, prompt sensitivity, biases, replicability limitations, homophily (AI-AI bias [16]), scalability risks, and incomplete cost accounting [14, 15, 17, 18]. We build on these critiques but argue that medical evaluation requires a stricter validation standard because medical outputs are often open-ended, context-dependent, and safety-critical, and may lack a single ground-truth answer [2, 3, 4, 9, 19]. Evaluating medical outputs therefore requires domain expertise: a judge may need to recognize clinically equivalent diagnoses, verify claims against clinical evidence, and determine whether generated notes are faithful to the source information [12, 13, 20]. To reduce

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the cost of clinician review, LLJs are increasingly used in clinical notes [2, 21], clinical diagnosis [22, 23], patient-facing QA [9], and treatment planning [4, 24].

To assess whether LLJs can supplement clinician evaluation, prior work has assessed medical LLJs by comparing their ratings with clinician ratings [25, 26]. Although such agreement is informative, it does not validate whether a judge measures the intended medical purpose, grounds its decisions in appropriate evidence, or produces consistent judgments across evaluation settings. Clinicians may also disagree when evaluation evidence is incomplete, multiple interpretations are defensible, or evaluation criteria are ambiguous [4, 9, 19, 27]. Interpreting LLJ–expert alignment therefore requires examining the clinical evidence, the clarity of the evaluation criteria, and the basis for clinician disagreement. In this paper, we revisit LLJ evaluation across clinical diagnosis, medical question answering, clinical-note generation, and treatment planning, identifying risks to reliability, clinical validity, and downstream safety that aggregate agreement alone can obscure. We further conduct a retrospective audit of MedHELM [4] and CQA-Eval [9] using outputs rated by multiple experts over downstream medical tasks. The audit shows substantial variation in LLJ–expert agreement across criteria, inconsistent scores across repeated runs, and sensitivity to evaluation design.

Our position is that LLJ–expert alignment should not be reported as standalone evidence of validity or safety. We propose a practical six-step protocol that integrates clinician expertise into evaluation design, validation, and targeted review while preserving the scalability of automated scoring. Four risk-proportionate readiness levels define minimum assurance requirements and use restrictions based on the intended use and the consequences of evaluation errors.

2 Related Work

Applications of LLJs in Medicine Beyond standard multiple-choice question (MCQ) benchmarks, medical evaluation increasingly includes open-ended generation tasks [12, 22, 28]. Prior work relied on automatic metrics such as ROUGE [29] and BLEU [30] to evaluate these tasks [31, 32, 33, 34]. However, lexical metrics are limited in medicine because they do not reliably capture clinical factuality, completeness, consistency, or safety [2, 35, 36]. Hager et al. [12] addressed this issue with a rule-based disease dictionary that evaluates free-text diagnoses against target pathology labels and accepts medically correct variants, such as "Perforated gallbladder" or "Gallbladder infection" for cholecystitis. However, such heuristic evaluation is difficult to scale across diseases, tasks, and criteria. Many studies relied on clinician evaluation [1, 4, 37], which is costly, slow, and hard to repeat at scale. This has motivated the use of LLJs as scalable substitutes or supplements to clinician review [7]. Recent research on medical reasoning evaluation suggested that LLJs are becoming a default component of medical NLG evaluation pipelines [4, 36, 38]. Specifically, Krolik et al. [7] showed that LLMs can serve as complementary evaluators alongside human expert ratings on Likert scales. Subsequent work has used LLJs to grade clinical notes [2], score medical summaries with fine-grained criteria [3, 39], judge whether predicted diagnoses match references [13, 23], rate harmfulness in dangerous medical questions [8], and support reasoning benchmark development, such as DiagnosisArena [40] and MedXpertQA [41]. At the same time, prior work showed that LLJ scores can be unstable [42, 43, 44], while human annotation is also not always a reliable gold standard [25]. Li et al. [45] further stated that grading-scale design can strongly affect LLJ performance: a 0–5 scale performed the best in most benchmark- and model-level comparisons. In HealthBench [19], researchers found that `GPT-4.1` exceeded the average physician score in most medical scenarios, with model–physician alignment roughly matching physician–physician agreement rate. However, their reported model–physician agreement was comparable to physician–physician agreement, supporting the alignment of model-based grading with expert judgment. However, such agreement does not necessarily establish clinical correctness, particularly when physicians disagree or when evaluation criteria are ambiguous. MedHELM [4] reported an intraclass correlation coefficient of 0.43 between clinicians and 0.47 between LLJs and clinicians, and concluded that LLJs may support procurement decisions. These findings motivate our central concern: LLJs are useful and scalable, but their reliability in higher-stakes medical evaluation remains underexplored.

Robustness of LLJs in Medical NLG Tasks Whether LLJs can be trusted as scalable evaluators in medicine depends on their robustness: do they produce stable evaluations across changes in prompts, judge models, repeated runs, grading scale, or evaluation criterion changes? Existing work suggested that robustness is task- and design-dependent, rather than a general property of LLJs.

Zhou et al. [46] evaluated LLJs for grading model-generated rationales against expert-written clinical reasoning trajectories in a medical QA setting. They found that LLJs were relatively stable across judge models and prompt variants when expert-written reference rationales were provided, especially with strong judges such as `GPT-4o mini` and `MedGemma-27B` [47]. Without reference rationales, the same judges became less stable. This shows that robustness of LLJs depends on judge models, prompt design, and the availability of high-quality references. Xie et al. [3] showed a similar pattern from the model-size perspective. Across SOAP-note generation, radiology-report summarization, and patient-question summarization, `GPT-4` substantially outperformed `Mistral-7B` as a judge. Several studies have used agreement with human raters as evidence of LLJ reliability. Brake et al. [2] reported Cohen's κ of 0.79, 1.00, and 0.32 between `Llama2-70B` and humans for age, gender, and body-part injury consistency in clinical note generation, respectively, showing reasonable alignment on objective criteria, but weak alignment on more subjective criteria. They also found that the prompting strategy mattered: one-shot prompting performed worst because the single example biased the judge. Li et al. [45] showed that LLJ panel reliability is sensitive to grading-scale choice and varies across evaluation criteria, with reliability dropping on long-context summarization tasks. This issue is especially important in medicine, where evaluation criteria such as factuality, completeness, safety, and clinical relevance are often interdependent. Vasilev et al. [10] further reported that, in medical summary evaluation, neither automatic metrics nor LLJ scores reliably detected factual errors and semantic distortions, even when LLJ relevance scores correlated with expert judgment. Overall, prior work has evaluated LLJ robustness using several strategies: varying judge models and prompts [3, 18, 46, 48]; measuring agreement with human raters using Cohen's κ or Krippendorff's α [2, 4, 19]; decomposing outputs into atomic claims rather than using only holistic scores [3, 49, 50]; and comparing reference-based with reference-free judging [46]. In this work, we discuss potential issues in using LLJs for medical tasks and argue that LLJ–expert alignment should not be reported as standalone evidence of validity or safety.

3 Risks of Using LLJs in Medical Evaluation

Medical LLJs are often justified by four practical promises: they reduce evaluation cost [26, 46, 51], produce stable scores [26, 46], align with clinicians [4, 9, 19, 52], and scale across medical tasks [3, 4, 26]. We argue that these promises are incomplete unless the evaluator is validated for the specific clinical task [14, 53] and the downstream decision it supports [12, 54, 55]. Across existing evidence and our case studies, we identify four risks inspired by prior literature: unstable medical judgments [26, 42, 56], opaque scoring [14, 15, 57, 58], weak validity under clinical uncertainty [4, 9, 19], and downstream safety and data-contamination risk [8, 12, 59, 60].

3.1 Risk 1: Unstable Medical Evaluation

An LLJ may give different ratings to the same medical output when the prompt, judge model, or scoring run changes. This variation makes it hard to tell whether score differences reflect the quality of the evaluated output or the evaluation setup. Prior work found that LLJ performance depended on the prompt and availability of reference information [2, 26, 46]. Closed-source LLJs were generally more robust to prompt design than open-source LLJs, especially with direct instructions, system roles, and detailed rubrics [26]. However, in multi-document clinical summarization, judge models showed inconsistent performance, and the same few-shot strategy helped some judges (e.g., `GPT-o3-mini`) but hurt others (e.g., `GPT-4o`) [56]. Prompt examples also affected agreement with human subjective, objective, assessment, and plan (SOAP) note ratings [2], and repeated calls can return different scores for the same answer even at temperature 0 [26]. These findings raise concerns about whether medical LLJs can produce stable ratings, even under the same evaluation setup.

To assess the repeatability of medical LLJ ratings, we evaluate three judges: `GPT-5.4 Nano` (Nano), `GPT-5.6 Luna` (Luna), and `Claude Sonnet 5` (Sonnet) on two datasets. MedHELM [4] provides model-generated clinical notes and treatment plans with paired human ratings for accuracy, completeness, and clarity. CQA-Eval [9] contains physician- and model-written answers to medical questions, each rated by three physicians for correctness, relevance, and risk disclosure. Both datasets use a 1–5 rating scale (the LLJ prompts are provided in App. D). For the same input and scoring prompt, as shown in Tab. 1, exact agreement between two calls ranges from 66.2% to 83.8% on MedHELM [4] accuracy criteria and from 67.0% to 75.2% on CQA-Eval risk disclosure. We also evaluate answer-order sensitivity using all 200 physician–model answer pairs from CQA-Eval [9], given each

Table 1: Exact self-agreement between two independent LLJ calls with identical model, data, and prompt settings. Numbers in parentheses give the number of evaluated items per task.

			Judge model		
Dataset	Task (n)	Criterion	GPT-5.4 Nano	GPT-5.6 Luna	Claude Sonnet 5
MedHELM	Clinical note generation (31) Treatment planning (8)	Accuracy	66.2%	83.8%	80.3%
CQA-Eval	Free-form medical QA (300)	Risk disclosure	67.0%	73.0%	75.2%

question has one physician answer and two model answers, each presented in both orders under two preference prompts. Across the six judge–prompt settings, reversing answer order changes the preferred answer directly from physician to model, or vice versa, on 11–40% of pairs. Nano switches on 40% of pairs under a prompt that explicitly asks the judge to ignore order. These findings show that LLJ ratings and answer preferences can vary without changing the evaluated content (App. B).

Table 2: Case study of LLJ (Luna) failures in diagnosis evaluation. Dx1 and Dx2 are drawn from MedCaseReasoning [23], and Dx3 is from MIMIC-CDM[a] [12]. The reasoning effort is set as default.

			LLJ judgment	
ID	Gold standard	Predicted diagnosis	Without system prompt	With system prompt
Dx1	Urticaria multiforme	Acute urticaria	✗ Incorrect	✓ Correct
Dx2	Oculomotor mononeuropathy	Post-vaccinal acute oculomotor nerve palsy	✓ Correct	✗ Incorrect
Dx3	Cholecystitis	Perforated gallbladder	✗ Incorrect	✗ Incorrect

[a] Dx3 appears in the rule-based evaluation rubric in the MIMIC-CDM supplementary materials. To protect patient privacy, we report only the final diagnoses, not the original case presentations.

Case Study: Clinical Diagnosis Evaluation Diagnosis evaluation shows why robustness in medicine requires more than stability in repeated scores. The task is to determine whether a generated diagnosis should be considered clinically equivalent to a reference diagnosis. This cannot be achieved through string matching alone. The same condition may have multiple aliases, disease variants may fall under the same diagnostic category, and clinically acceptable answers may differ in specificity. Therefore, an LLJ must apply task-specific medical equivalence rules, not only judge semantic similarity. As shown in Tab. 2, the LLJ (Luna) produces false-negative judgments in diagnosis accuracy evaluation using the prompt from MedCaseReasoning [23] (see App. D.1). A false-negative judgment in this setting rejects a diagnosis that should be accepted under the evaluation policy, potentially underestimating the diagnostic performance of the evaluated model. Prior work commonly used ICD-10 codes or rule-based matching to evaluate diagnostic correctness [12, 61]. In Dx1, "Urticaria multiforme" is a clinical subtype of "Acute urticaria," and are judged as "Incorrect." Notably, the LLJ changes from "Incorrect" to "Correct" after adding a simple system prompt: *"You are a professional medical expert that answers only 'y' or 'n'."* In Dx2, both diagnoses fall under the same ICD-10 category, H49.0 (third [oculomotor] nerve palsy), but should not be treated as equivalent because post-vaccinal palsy is one rare cause of oculomotor mononeuropathy, not another name for it. In Dx3, the MIMIC-CDM rule-based evaluation treats "Cholecystitis" and "Perforated gallbladder" as the same pathology [12], but the LLJ labels the pair as incorrect. These examples show that LLJ robustness is not only about score stability. A judge can be consistently wrong when the prompt does not encode task-specific medical rules.

Prompt design can reduce LLJ errors, but it does not fully solve the problem. Qiu et al. [13] proposed a more detailed diagnosis-evaluation prompt that includes examples and explicitly states that the same disease may have multiple aliases. Applying this prompt to the cases in Tab. 2 corrects Dx1 and Dx2, but Dx3 remains incorrectly labeled (see prompt in App. D.1). This suggests that better prompts can improve LLJ behavior, but they cannot fully solve medical equivalence without stronger grounding in clinical rules and references. This concern aligns with Krumdick et al. [62], who found that LLJs aligned highly with experts only when the judge could correctly answer the question, especially in knowledge-intensive settings. Judge model choice is another source of instability. In multi-document clinical summarization, smaller open-source judges aligned poorly

with expert annotations: `Llama 3.1 8B` achieved an agreement rate of 33.2%, while `GPT-o3-mini` reached 81.8% [56]. Supervised fine-tuning improved `Llama 3.1 8B` to 56.0%, but still did not match stronger proprietary judges [56]. Reliable medical judges require explicit evaluation rules, appropriate references, and judge models capable of applying them consistently.

To examine how task-specific acceptance rules introduce another source of variation, we evaluate 160 diagnosis predictions generated by `DeepSeek-V4-Flash` [63], comprising 80 randomly selected MedCaseReasoning cases [23] and all 80 MIMIC-CDM cases [12]. Changing the evaluation rule from matching the full ICD-10 code to matching its first three characters increases measured accuracy by 18.8 percentage points without changing the predictions. This sensitivity analysis shows that reported diagnostic accuracy depends on the chosen acceptance rule, not as evidence that either matching rule is universally superior.

High accuracy under a single judge–prompt configuration does not establish reliable medical evaluation; the judge, prompt, and scoring rules should be validated together, with robustness tested across models, prompts, and repeated runs.

3.2 Risk 2: Opaque Scores Hide Clinically Meaningful Errors

Prior work has warned that aggregated evaluator scores can hide what is actually being measured [6, 64]. Measurement-theoretic critiques of NLG evaluation have argued that metric scores are difficult to interpret when the target construct is underspecified [14], and recent LLJ studies similarly raised concerns about opaque judge behavior, unreliable rationales, and limited validity of model-based scoring [15, 57]. Medical NLG makes this problem sharper because the score is often used to judge hidden clinical qualities, such as whether a diagnosis is acceptable [12, 13], whether a summary preserves important evidence [3, 10, 11, 65, 66], or whether a treatment plan is safe [4]. Fine-grained medical evaluation frameworks, such as DocLens, decomposed outputs into atomic claims precisely because holistic scores can obscure whether an error is an omission, unsupported claim, attribution failure, or distortion of clinical meaning [3]. Studies in medical summarization further showed that automatic metrics and LLJ scores can correlate with expert judgments while still failing to reliably detect factual errors and semantic distortions [10, 11]. Thus, a single LLJ score can hide why the judge accepted or rejected an answer, and whether the hidden error is clinically important.

Table 3: Agreement on the multi-annotator MedHELM subsets. For each judge, the two values give agreement with two human annotators. The final column averages the three judges and both annotators across the three criteria.

MedHELM Tasks	Human–Human	GPT-5.4 Nano	GPT-5.6 Luna	Claude Sonnet 5	Mean LLJ
Clinical notes ($n = 31$)	37.6%	38.7% / 29.0%	45.2% / 37.6%	51.6% / 41.9%	40.7%
Treatment plans ($n = 8$)	33.3%	29.2% / 4.2%	20.8% / 16.7%	16.7% / 29.2%	19.4%

Case Study: Clinical Note Evaluation We audit all 31 notes and 8 treatment plans in its selected multi-rater subsets from MedHELM [4]. Each output has two human ratings. Mean exact judge–human agreement across Nano, Luna, and Sonnet is 40.7% for clinical notes, compared with 37.6% human–human agreement. For treatment planning, the corresponding values are 19.4% and 33.3% (Tab. 3). For note completeness, agreement with individual human raters ranges from 12.9% to 45.2%, compared with 51.6% to 67.7% for clarity. The judges therefore reproduce clarity scores more frequently than completeness scores in this sample (App. A.1). An example case in Tab. 11 shows that LLJ gives the generated note from `Gemini 1.5 Pro` [67] perfect scores on all three criteria, while the human expert gives lower scores for accuracy and completeness. A fluent note can therefore receive favorable judge scores without preserving all relevant source information.

Aggregate agreement can conceal failures on clinically important criteria; reliable medical evaluation requires criterion-level validation and source-grounded analysis of the errors that LLJ scores detect or miss.

3.3 Risk 3: Weak Validity Under Clinical Uncertainty

Medical LLJs are often validated by comparing their scores with clinician ratings [4, 19]. This comparison also depends on whether clinicians use the rating criteria consistently. Agreement with inconsistent clinician ratings cannot provide a clear reference for LLJ validity. We therefore examine how LLJ agreement varies with clinician consensus and compare it with physician–physician agreement for each criterion. In our analysis of 300 CQA-Eval answers rated by three physicians [9], pairwise exact agreement between physicians is 45.3% for correctness, 47.2% for relevance, and 26.3% for risk disclosure (Tab. 4). Using the `GPT-4o` judge ratings from the original study, we find that even among items with unanimous physician scores, exact judge–physician agreement reaches 98.6% for correctness ($n = 216$ rating comparisons), but only 52.2% for risk disclosure ($n = 69$). Physician disagreement can reflect incomplete evidence, different interpretations of a rating criterion, or annotation noise. The results show that agreement must be examined by criterion and alongside physician–physician agreement, rather than treated as proof that an individual LLJ score is correct.

Case Study: Free-form Medical QA Evaluation Tab. 5 shows three CQA-Eval answers scored for correctness, relevance, and risk disclosure. QA1 shows a disagreement about relevance, and QA2 shows a disagreement about risk disclosure. In QA3, the judge assigns a perfect correctness score to an answer stating that there is no evidence that allopurinol directly affects fertility. Although animal studies do not show impaired fertility at the tested doses, the FDA label lists male infertility among reported adverse reactions[2]. A reported adverse reaction does not establish causality, but the answer's unqualified statement does not reflect all available evidence. The example illustrates why correctness requires checking the strength and limits of the cited evidence, rather than relying on a favorable judge score.

Table 4: Exact agreement of original LLJ `GPT-4o` with physician ratings on CQA-Eval [9]. Parentheses are the number of ratings. The final column reports pairwise physician exact agreement.

	Physician consensus		
Criterion	Unanimous	All differ	Human–human
Correctness	98.6% (216)	32.4% (108)	45.3%
Relevance	90.9% (231)	33.3% (87)	47.2%
Risk disclosure	52.2% (69)	25.1% (327)	26.3%

Table 5: Case study of LLJ (original `GPT-4o` ratings) failures in free-form medical QA (CQA-Eval [9]). Scores are correctness (C), relevance (R), and safety (S) on a 1–5 Likert scale. Each cell shows the average (avg.) of three physician ratings, with the LLJ score in brackets below.

			Expert Avg. [LLJ]		
ID	Question	Answer	C	R	S
QA1	What is meningitis?	*[Llama 3.1 405B]* Treatment depends on the underlying cause, but may include antibiotics… to manage symptoms and prevent complications.	4.67 [5]	1.00 **[4]**	1.00 [1]
QA2	How much melatonin should I take at night?	*[Physician]* For chronic sleep disorders, consult a doctor or sleep specialist for diagnosis and consider treatment such as cognitive behavioral therapy.	5.00 [5]	2.00 [2]	4.00 **[1]**
QA3	Does allopurinol affect fertility?	*[GPT-4]* There is no evidence to suggest that allopurinol has a direct effect on fertility…	3.67 **[5]**	3.67 **[5]**	1.33 [1]

To further examine the frequency of LLJ–expert disagreement beyond these examples, we compare answer-level LLJ and median physician risk-disclosure ratings using a threshold of 4 on CQA-Eval (Tab. 6). `GPT-4o` more often assigns favorable scores to answers with low physician ratings, while Nano, Luna, and Sonnet more often assign low scores to answers rated favorably by physicians. These patterns raise concerns about the validity of LLJs, as judges may reward answers that omit treatment risks or contraindications and give low scores to answers that physicians consider to explain them, requiring clinical review to distinguish missed clinical information from differences in interpretation (see App. A.2 for additional audit on other potential agreement discrepancies).

Case Study: Treatment Planning Evaluation In Tab. 12, `GPT-4o` assigns a lower completeness score than both clinicians (3 vs. 5) to a treatment plan, overlooking a clinical constraint that limits

[2] `https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=10cc918f-aa44-415b-932d-2404695ac449` and `https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=30ad5ba2-1cef-4933-b104-0597f6a2aaa2`

Table 6: Risk-disclosure flags for each judge, defined against the median of three physician ratings on CQA-Eval [9]. Each cell gives the number of flagged answers, with the percentage of the subset in parentheses. These discrepancies were not independently adjudicated as clinical errors.

	Condition			Judge model			
Directional Risk-Disclosure Flag	Physician	Judge	n	Nano	Luna	Sonnet	GPT-4o
Potential missed disclosure	≤ 3	≥ 4	138	14 (10%)	22 (16%)	3 (2%)	66 (48%)
Potential false alarm	≥ 4	≤ 3	162	102 (63%)	97 (60%)	114 (70%)	10 (6%)

the feasibility of bypass surgery. Across the eight treatment plans, the three newer judges match individual human scores in 19.4% of comparisons on average, compared with 33.3% agreement between the two human raters (Tab. 3). These findings highlight that LLJ validity should not be assumed to generalize across medical tasks.

LLJ–expert agreement must be interpreted in light of uncertainty in the clinical reference; establishing LLJ validity requires examining expert disagreement, weighing the underlying evidence, and adjudicating clinically consequential discrepancies.

3.4 Risk 4: Downstream Safety and Data Contamination

LLJ ratings can affect which medical models are selected, which generated examples are retained for training, and which outputs receive further review [14, 15, 59]. The CQA-Eval threshold analysis shows that switching the judge changes how many answers are flagged for risk-disclosure review (Tab. 6). The diagnosis experiment shows that changing the acceptance rule changes measured accuracy by 18.8 percentage points without changing the predictions (Sec. 3.1). Neither experiment measures a downstream clinical outcome or a completed model-selection decision. They show how the evaluation setup can change the evidence used to make such decisions. If LLJ scores are used to filter synthetic clinical notes or patient–physician conversations, similar errors may cause incomplete or unsupported examples to remain in a training dataset [68, 69]. Prior work supports this concern. Vasilev et al. [10] found that automatic metrics and LLJ scores did not reliably detect factual errors and semantic distortions, even when some LLJ scores correlated with expert judgment. Tang et al. [11] similarly showed that standard automatic metrics failed to capture factual inconsistency, potential medical harmfulness, and human preference in medical evidence summarization. Therefore, LLJs have not yet been shown to be safe replacements for clinician review in medical tasks.

Nevertheless, LLJs address a genuine limitation of medical evaluation. Clinician annotation is expensive and difficult to repeat at the scale of current benchmarks, while automated scoring allows researchers to compare many outputs consistently when the judge has been validated for the task [4, 56]. Stronger reasoning models, retrieval, and agentic workflows may also improve medical judging. However, retrieving a reference does not establish that the source is up-to-date, supports the claim, or applies to the patient and clinical setting. A judge may also share errors with the model it evaluates [14, 15]. These developments can reduce particular failure modes, but they do not replace criterion-level validation or clinical review of high-severity disagreements. A convincing claim that LLJs can replace clinician review should report the number and type of clinician annotations avoided, uncertainty across prompts and runs, clinical failure modes missed by the LLJ, and whether model-selection or data-filtering decisions change when the evaluator changes [14, 15, 54]. We argue that current reporting practices for medical LLJs are insufficient: judge–expert alignment should not be reported as standalone evidence of validity or safety.

When LLJ scores guide model selection or synthetic-data filtering, evaluation errors can propagate into future models and clinical decisions; validation and clinical oversight should therefore scale with the consequences of those uses.

4 A Protocol for Using LLJs in Medical Evaluation

We argue that medical LLJs should not be used as default evaluators without task-specific validation [14, 53]. We propose a six-step protocol that uses clinical expertise for defining and validating

an evaluator rather than requiring clinicians to score every model output. Clinicians develop a task-specific codebook, annotate a calibration sample, and adjudicate uncertain or potentially consequential disagreements. An LLJ can evaluate the remaining outputs only within the scope supported by that validation. The protocol draws on human evaluation and medical AI reporting guidance [24, 70, 71, 72]. Tab. 7 specifies proposed requirements for different uses of LLJ scores.

4.1 Protocol Design

Step 1: Define the evaluation construct and codebook. Before prompting an LLJ, researchers should work with domain experts to define quality for the target task. The codebook should specify dimensions such as diagnostic equivalence [12, 13], source faithfulness and factuality [3, 10, 11, 73], safety and completeness [4, 9], clinical appropriateness, clarity, or patient comprehension. Each dimension should include a definition, rating scale, examples, exclusion criteria, and guidance on whether judgments are primarily objective or subjective [14, 24]. For diagnosis evaluation, specify how to handle aliases, disease variants, ICD-10 specificity, and parent-child categories [12, 13, 61]. For clinical notes, define unsupported findings, omitted relevant negatives, and acceptable paraphrases [3, 4]. This step targets Risk 1 (Sec. 3.1) by turning implicit clinical judgment into explicit decision rules.

Step 2: Build a human-annotated validation set. Before scaling, researchers should use random or stratified sampling to select model outputs for expert annotation. The sample should cover different output qualities, task types, clinical topics, and expected edge cases [24, 70]. In medical NLG, include diagnosis aliases, partially correct drug-safety claims, unsupported or incomplete clinical notes, and treatment plans whose source information rules out an otherwise plausible recommendation. At least two qualified experts should independently annotate each item using the codebook. Report inter-annotator agreement for each dimension and major sources of disagreement. Clinically meaningful disagreements should be adjudicated into a reference label or an explicit "ambiguous" category. This step addresses Risk 3 (Sec. 3.3) by making the human reference standard explicit instead of assuming that one expert label is ground truth.

Step 3: Design the LLJ prompt from the human codebook. The LLJ prompt should follow the human codebook's dimensions, scales, and decision rules. It should define the task, criteria, output format, and allowed evidence, such as the source note, reference answer, diagnostic taxonomy, drug label, or clinical guideline [3, 9, 53, 74]. Require criterion-specific judgments instead of vague instructions (e.g., "judge whether the answer is good"). Diagnosis prompts should address aliases and specificity, note-generation prompts should require source support for accepted findings, and patient-facing QA prompts should separately score correctness, relevance, or safety disclosure. Where possible, LLJs should cite the source span, rule, or clinical reference supporting each score. Avoid using models from the same family as the generator and judge as this may introduce shared blind spots or self-preference bias [53]. This step addresses Risk 2 (Sec. 3.2) by linking each score to a clear criterion and evidence.

Step 4: Validate LLJ–expert alignment before scaling. LLJ–expert alignment should not be reported as standalone evidence of validity or safety. Evaluate judges on an expert-rated validation set before benchmarking or model selection. Of the 42 studies reporting quantitative judge–human validation in our structured review, 20 omit human–human reliability, limiting interpretation of judge agreement relative to reference consistency. Report judge–human and human–human agreement for each criterion, and test robustness across repeated runs, prompt variants, and multiple judge models [2, 26, 42, 56, 75]. The Alternative Annotator Test [76] can complement descriptive agreement measures by comparing LLJs with individual annotators relative to the remaining panel. In medicine, passing such a test should be accompanied by clinical error analysis and validation for the intended downstream use. Examine performance across objective and subjective criteria, common and rare cases, high-risk and low-risk outputs, and fluent but clinically flawed responses. For higher-risk uses, also test position/order effects, format or scale changes, verbosity bias, demographic/context variants, and correlated errors across same-family judges [5, 44, 53, 77]. Calibrate any confidence scores against expert judgments and use them only as auxiliary uncertainty signals. For consequential uses, assess whether evaluation changes alter model rankings or filtering decisions. This step addresses Risk 1 (Sec. 3.1) and Risk 4 (Sec. 3.4) by testing whether LLJ scores are stable enough for the intended downstream use.

Step 5: Conduct clinical error analysis. Clinicians should review disagreements and classify confirmed errors by task-specific criteria [10, 11, 12]. Categories should include missed safety issues [8], unsupported source claims [3, 10, 78], clinically important omissions [4], over-penalized diagnosis aliases [13], incorrect use of ICD specificity [12, 61], overvalued fluency [11], or failure to apply a clinical rule [62]. Assess both error frequency and acceptability for the intended use. A higher LLJ score than a physician score does not by itself establish a missed clinical error. Check the source material and conflicting clinician ratings before labeling a case. Analyze high-severity discrepancies, and limit judge use for model ranking [14], procurement decisions [4], or deployment justification [12, 53, 79] if failures remain unresolved. This step connects validation to Risk 4 (Sec. 3.4), where evaluator errors may contaminate synthetic-data pipelines or distort model-selection decisions (see App. A.1 for detailed audit results).

Step 6: Report and release evaluation artifacts. Studies should report the codebook, expert annotation procedure, annotator qualifications, inter-annotator agreement, adjudication process, LLJ prompt, judge model, sampling strategy, prompt variants, run count, LLJ–expert alignment report, and clinical error analysis [53, 70, 80, 81]. Report the LLJ's intended role and risk level, including preliminary triage, benchmarking, synthetic-data filtering, model ranking, procurement support, or deployment justification. For systems that affect clinical decisions, include version control, revalidation plans, and audit artifacts [53, 82]. Where possible, release the human-annotated validation set, codebook, prompts, and evaluation scripts. These artifacts support interpretation, reproducibility, and comparison across studies. This step addresses Risk 2 (Sec. 3.2) and Risk 4 (Sec. 3.4) by making downstream use, validation scope, and evidence behind scores explicit.

Table 7: Risk-proportionate LLJ readiness levels. The evidence required depends on the intended use and on the consequences of an incorrect evaluation.

Level	Permitted LLJ use	Minimum assurance and restriction
L0 Exploratory	Error analysis, prompt comparison, hypothesis generation.	Report criteria, prompt, judge version, settings, and run-to-run variability; state that the judge is unvalidated. No safety, superiority, or deployment claims.
L1 Validated reuse	Reuse within a benchmark's validated scope, or a low-risk task that is easy to check.	Confirm that task, population, criterion, judge, and prompt are still in scope; report repeatability; add expert review if the scope or conclusion changes.
L2 Benchmark / model selection	Building benchmarks, ranking models, release decisions.	Expert codebook; multi-clinician reference on a sample; pre-specified margin against expert–expert agreement; prompt, run, and judge stress tests; $\geq$95% bootstrap winner stability; adjudication of high-severity cases.
L3 Real-world use	Procurement, deployment, workflow decisions, safety-critical filtering.	Everything in L2, plus prospective or external validation, audit trails, monitoring, revalidation, and human override. Unresolved high-severity failures block use.

4.2 Practical feasibility and readiness levels

Prior studies demonstrate the operational feasibility of validated LLJs at scale: Croxford et al.'s judge achieved an Intraclass Correlation Coefficient (ICC) of 0.818, with mean evaluation time decreasing from 600 seconds for physician review to 22 seconds for automated evaluation at USD 0.05 per clinical summary [56]. Similarly, LLMEval-Med evaluated 13 models on 2,996 questions [83]; MedHELM engaged 29 clinicians across 121 tasks for nine models and 35 benchmarks [4]; and HealthBench utilized 262 physicians to generate 48,562 criteria across 5,000 conversations [19]. These findings establish that expert-curated criteria enable reliable scaling.

However, LLJs do not inherently reduce total evaluation costs. Full clinician evaluation of N outputs with k ratings each requires Nk ratings. Our approach, by validating a subset of m outputs, applying the LLJ to the remainder, and referring a fraction e back for human review, requires $mk + e(N-m)k$ ratings, reducing human effort by $(N-m)(1-e)k$. Nonetheless, this introduces expenses for LLJ inference, prompt testing, and monitoring. Croxford et al.'s USD 50 physician-labor and USD 0.05 API estimates cover distinct expense types rather than full validation costs [56]. Feasibility thus does not guarantee cost savings. Future evaluations should report clinician hours, reviewed proportions, LLJ inference costs, latency, and their effect on model rankings (see App. C for the full cost model).

We further propose four readiness levels according to how LLJ scores are used (Tab. 7). **L0** permits exploratory analysis with transparent settings; **L1** allows reuse within a previously validated scope; **L2** requires expert validation, robustness testing, and ranking-stability checks for benchmarking or model selection; and **L3** adds external validation and human oversight for real-world deployment. Requirements scale with intended use rather than study context. The proposed numerical gates are provisional starting points rather than universal standards.

4.3 Using LLJs From Prior Work

When reusing an LLJ prompt, judge model, or evaluation framework, researchers should first check whether the original study included human validation, expert annotation, inter-annotator agreement, prompt-robustness testing, bias testing, and clinical error analysis. Frameworks such as MedJUDGE can guide this checklist, but they do not automatically validate a new task, judge model, or risk level [53]. Researchers should validate the LLJ on a task-specific expert-annotated subset before scaling. They should also check whether the prompt covers the intended construct: general helpfulness is not enough to measure clinical safety, source faithfulness, diagnostic equivalence, treatment appropriateness, or patient-specific risk disclosure.

4.4 Counterarguments

Scalability and efficiency remain important advantages of LLJs, and appropriately validated judges may perform well within a defined scope. Advances in reasoning, retrieval, memory, and agentic workflows may further reduce current failures by helping judges search and verify medical evidence. Greater capability or access to external sources does not by itself establish clinical validity because a judge may retrieve outdated evidence, apply it to the wrong clinical setting, or share blind spots with the model it evaluates. Our position does not advocate replacing LLJs with exhaustive clinician review. Instead, clinicians can focus on developing evaluation criteria, validating representative subsets, and analyzing consequential disagreements, while validated LLJs score the remaining outputs. Among the 42 prior studies reporting judge–human validation, 20 did not report human–human reliability. These limitations support task-specific codebooks, multiple clinician references, and clinical error analysis rather than treating either a stronger judge or agreement with a single clinician as sufficient evidence of LLJ validity.

5 Conclusion and Future Directions

Our proposed protocol highlights several priorities for future medical LLJ research. Rather than relying on aggregate expert agreement, studies should validate LLJs for specific tasks and evaluation criteria, define intended uses, and account for applicable regulations, which may vary across jurisdictions and evolve over time. Researchers should develop clear codebooks and expert validation sets with clinically meaningful edge cases, and design LLJ prompts based on these codebooks [14, 24, 53]. They should also examine how prompts, repeated runs, grading scales, and judge models affect evaluation results before applying LLJs at scale [2, 26, 42, 45, 56]. To support reproducibility, studies should report judge–expert agreement, clinical error analysis, intended use, and key evaluation details, including codebooks, prompts, judge versions, validation sets, and disagreement analyses [70, 71, 80, 81]. Without such validation and reporting, LLJ scores should not be treated as reliable measures of medical model quality across different settings.

This paper argues that medical LLJs should be treated as task-specific evaluation approaches, not automatic substitutes for clinician review. Although LLJs can reduce human annotation burden and support large-scale evaluation, current evidence does not show that they are consistently stable, transparent, clinically valid, or safe across medical tasks. Our audit finds differences in agreement across multiple medical tasks, unstable ratings across various judges, answer-order sensitivity in expert–model comparisons, and substantial differences in subjective evaluation ratings. Changing the diagnosis acceptance rule also changes measured accuracy without changing model predictions. These failures matter because LLJ scores can shape benchmark conclusions, model selection, synthetic-data filtering, and downstream clinical trust. Responsible use requires task-specific codebooks, expert validation, human-derived prompts, robustness testing across prompts, models, clinical error analysis, and transparent reporting of artifacts and intended use. LLJs may be useful, scalable supplements, but in medicine, they should remain validated aids to expert evaluation, not universal evaluators.

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A Retrospective Audit: Agreement and Clinical Discrepancies

A.1 Audit Design and Agreement with Clinicians

To provide operational evidence of our proposed protocol (Sec. 4), we conduct a retrospective audit on Steps 4-5 using clinician-rated outputs from MedHELM [4] and CQA-Eval [9]. The multi-rater MedHELM subset contains 31 clinical notes and eight treatment plans, with two human ratings per output. One of the clinical-note annotators is a medical student. CQA-Eval contains 300 answers, each rated by three physicians. We evaluate these outputs with `GPT-5.4 Nano` (Nano), `GPT-5.6 Luna` (Luna), and `Claude Sonnet 5` (Sonnet). We also analyze the original `GPT-4o` judge ratings of CQA-Eval. All scores use a 1–5 scale, with higher scores indicating more favorable evaluations. We use *risk disclosure* for the CQA-Eval criterion about communicating risks and contraindications.

We report *exact agreement*, meaning that two ratings have the same numerical score. In MedHELM, each judge score is compared separately with both human scores, giving 62 note-rating and 16 treatment-plan-rating comparisons per criterion. Human–human agreement uses one comparison per output. For CQA-Eval, agreement with each of the three physicians is reported separately. Human–human agreement serves as a reference, not as proof that any individual rating is correct.

Tab. 8 provides criterion-level agreement with each human annotator for clinical-note generation and treatment planning. For clinical notes, judge–human agreement ranges from 12.9% to 45.2% for completeness, compared with 51.6% to 67.7% for clarity. These differences are not reflected in the task-level averages reported in Sec. 3.2. The treatment-planning results are descriptive because the subset contains only eight outputs.

Table 8: Criterion-level exact agreement on MedHELM. For each judge, values indicate exact agreement with Annotator 1 and Annotator 2, respectively. All values are percentages.

Criterion	Human–Human	GPT-5.4 Nano	GPT-5.6 Luna	Claude Sonnet 5
Clinical-note generation ($n = 31$)				
Accuracy	41.9	29.0 / 22.6	35.5 / 29.0	45.2 / 41.9
Completeness	22.6	29.0 / 12.9	38.7 / 19.4	45.2 / 16.1
Clarity	48.4	58.1 / 51.6	61.3 / 64.5	64.5 / 67.7
Treatment planning ($n = 8$)				
Accuracy	62.5	12.5 / 12.5	0.0 / 25.0	25.0 / 50.0
Completeness	25.0	37.5 / 0.0	25.0 / 0.0	25.0 / 0.0
Clarity	12.5	37.5 / 0.0	37.5 / 25.0	0.0 / 37.5

We also examine whether `GPT-4o` agreement changes with physician consensus. For each CQA-Eval criterion, we compared answers for which all three physicians gave the same score with those for which all three gave different scores. The numbers in parentheses in Tab. 4 count physician–judge rating comparisons, rather than independent answers. The unanimous strata contain 72, 77, and 23 answers for correctness, relevance, and risk disclosure, respectively. The all-different strata contain 36, 29, and 109 answers. Answers on which exactly two physicians agreed are excluded from this contrast.

`GPT-4o` agreement is lower when all three physicians disagree, particularly for risk disclosure. This association does not show that disagreement causes judge errors or that every disagreement reflects clinical uncertainty. Physicians may differ because of ambiguous evidence, different interpretations of the rating criteria, or annotation noise. Evaluation reports should therefore show the distribution of expert ratings and whether disputed items are adjudicated.

A.2 Directional Discrepancies and Illustrative Cases

We screen for score differences that may deserve clinician review. These rules identify *potential discrepancies*, not confirmed clinical errors. In the MedHELM note subset, each item can contribute up to two judge–human comparisons ($N = 62$ per criterion). The flags in Tab. 9 can overlap. A higher judge accuracy score does not itself prove that an unsupported claim is accepted, nor does a higher completeness score prove that an omission is missed. Such judgments require inspection of the source and generated note. For CQA-Eval, 420 individual physician ratings of risk disclosure are

at most 3. For example, GPT-4o assigns a higher score in 365 comparisons and a score of at least 4 in 227. These are rating-level counts and are not directly comparable with answer-level counts.

Table 9: Directional screening counts for each judge model. J is the judge score and H is an individual physician score. Subscripts A, Co, and Cl stand for accuracy, completeness, and clarity, and RD stands for risk disclosure. MedHELM counts are out of 62 note-rating comparisons. CQA-Eval counts are out of the 420 physician ratings with $H_{RD} \leq 3$. The highest count in each row is shown in bold.

		Judge model		
Screening flag	Condition	Nano	Luna	Sonnet
MedHELM (N=62)				
Fluency over substance	$J_{Cl} \geq H_{Cl} \wedge (J_A < H_A \vee J_{Co} < H_{Co})$	**42**	41	37
Unsupported claim accepted	$J_A > H_A$	11	13	**18**
Omission missed	$J_{Co} > H_{Co}$	9	10	**13**
CQA-Eval, risk disclosure (N=420, $H_{RD} \leq 3$)				
Risk overrated	$J_{RD} > H_{RD}$	158	**213**	138
Inadequate risk rated adequate	$J_{RD} \geq 4$	58	**77**	26

At the answer level, we use the median of each answer's three physician scores. There are 138 answers with a median risk-disclosure score of at most 3 and 162 with a median score of at least 4. Tab. 6 shows judge scores that cross these thresholds in either direction. A potential missed disclosure may reward an answer with incomplete warnings; a potential false alarm may penalize an adequately cautious answer or add review work. The different directions across judges matter. GPT-4o has more potential missed disclosures in this analysis, while Nano, Luna, and Sonnet have more potential false alarms. The counts motivate targeted expert review; they do not establish which judge is safest.

The CQA-Eval example asking whether miconazole powder is the same as regular baby powder illustrates why criterion-level inspection matters. The GPT-4 answer distinguishes an antifungal medicine from an ordinary moisture-absorbing powder and advises appropriate use, but does not address inhalation or aspiration risks or when to seek care. In Tab. 10, GPT-4o matches the physician median on correctness and relevance, yet rates risk disclosure three points higher. This discrepancy merits review without proving that the judge made a clinically confirmed error.

Table 10: Ratings of the miconazole-powder answer from three physicians and the GPT-4o judge, on a 1–5 scale where higher is better. The judge matches the physicians on correctness and relevance but rates risk disclosure 3 points above the physician median. Confidence was collected from physicians only.

	Score (1–5)			
Rater	Correctness	Relevance	Risk disclosure	Confidence
Physicians				
Physician 1	4	4	2	Fairly confident
Physician 2	5	5	1	Very confident
Physician 3	5	5	1	Very confident
Median	5	5	1	–
GPT-4o (LLJ)	5	5	4	–
Δ (LLJ − median)	0	0	+3	

As shown in Tab. 11, the LLJ gives the generated note from Gemini 1.5 Pro [67] perfect scores on all three criteria, while the human expert gives lower scores for accuracy and completeness. The generated note includes unsupported information, such as shortness of breath occurring "even at rest," which is not clearly stated in the source conversation. It also misses relevant negatives and contextual details, including no hematemesis, no bleeding elsewhere, no bruising or gum bleeding, and no recent emergency-room visit. This failure is clinically important because unsupported findings and missing negatives can change how a clinician interprets the case. Overall, these examples show that LLJ–expert alignment is an incomplete measure of validity. LLJs may align with clinicians on

Table 11: Case study of an LLJ–human disagreement in clinical note generation (MedHELM [4]; LLJ is the original `GPT-4o`). Scores are accuracy (Acc), completeness (Co), and clarity (Cl) on a 1–5 Likert scale. Each cell shows the expert rating, with the LLJ score in brackets below.

			Expert [LLJ]		
ID	Clinical conversation	Generated clinical note	Acc	Co	Cl
CNG1	*Summarize the conversation to generate a clinical note with four sections: 1. History of present illness; 2. Physical exam; 3. Results; 4. Assessment and plan.* [doctor] hi alan, how are you? [patient] hi, good to see you. [doctor] so alan, what's going on? [patient] so over the weekend, uh, i took a very long bike ride, about three hours or so and i got back, i was very light headed, i was short of breath and it really surprised me because i pretty much bike ride every morning and it was never an issue before so i'm a little concerned about that. [doctor] okay, and have you had any other symptoms? have you had any fever? [patient] no [doctor] chest pain? [patient] no, but i felt very light headed and…	**Clinical Note** **Patient:** Alan **Age:** 61 **Date:** Today's Date **1. HISTORY OF PRESENT ILLNESS** This is a 61-year-old male presenting for evaluation of shortness of breath… **2. PHYSICAL EXAM** **Vitals:** All vital signs within normal limits… **3. RESULTS** – Chest x-ray: Normal, no pulmonary edema or pneumonia… **4. ASSESSMENT AND PLAN** Based on the patient's history, exam, and test results, the likely diagnosis is an asthma exacerbation potentially triggered by exercise and/or allergies…	3 [5]	3 [5]	4 [5]

some objective criteria, but they can still miss clinically meaningful errors, overrate unsafe claims, or reward fluent but unsupported outputs.

Table 12: Case study of an LLJ (original `GPT-4o` ratings) failure in treatment planning (MedHELM [4]). Scores are accuracy (Acc), completeness (Co), and clarity (Cl) on a 1–5 Likert scale. Each cell shows the expert rating, with the LLJ score in brackets below.

			Expert [LLJ]		
ID	Task input	Predicted answer	Acc	Co	Cl
TP1	*Given various information about a patient, return a reasonable treatment plan for the patient.* Medical specialty: Cardiovascular / Pulmonary Sample name: Aortic valve replacement Description: Aortic valve replacement using a mechanical valve and… Diagnosis: Aortic valve stenosis with coronary artery disease associated with congestive heart failure… Findings: The left ventricle is certainly hypertrophied… Keywords: cardiovascular / pulmonary, … Answer: …	Based on the provided medical information, a reasonable treatment plan for the patient can be outlined as follows: **Short-term treatment plan:** 1. **Post-operative care**: The patient will require close monitoring in the intensive care unit (ICU) for at least 24–48 hours… 2. **Pain management**: The patient will require adequate pain control… **Long-term treatment plan:** 1. **Lifestyle modifications**: The patient will be advised to make significant lifestyle changes, including: … 2. **Medication management**: The patient will require long-term medication management…	4 [4]	5 [3]	3 [5]

Tab. 12 presents a treatment plan generated by `Llama 3.3 70B Instruct` [84]. Both clinicians assign a completeness score of 5, whereas `GPT-4o` assigns 3 and asks for more discussion of diabetes and obesity management. The source states that severe obesity limits the available conduit and that bypass is not attempted. The judge's criticism does not fully account for this constraint.

Tab. 13 also shows the individual human ratings for the clinical-note (CNG1) and treatment-plan (TP1) examples. Both annotators give CNG1 identical ratings across all three criteria. For TP1, the clinicians agree on accuracy and completeness but differ on clarity.

Table 13: Individual human ratings and GPT-4o scores for examples CNG1 and TP1 (1–5 scale). The two human raters agree exactly on 3/3 criteria for CNG1 and on 2/3 for TP1; these fractions describe only the examples shown.

		Score (1–5)		
Example	Rater	Accuracy	Completeness	Clarity
CNG1	Annotator 1	3	3	4
	Annotator 2	3	3	4
	GPT-4o	5	5	5
TP1	Clinician 1	4	5	3
	Clinician 2	4	5	5
	GPT-4o	4	3	5

B Robustness to Judge Settings and Evaluation Rules

B.1 Prompt Choice and Answer Order

We assess preference judgments using 200 physician–model answer pairs from 100 CQA-Eval questions. Each physician answer is paired with one answer generated by `GPT-4` and one generated by `Llama-3.1-405B`. Presenting every pair in both answer orders produced 400 choices per judge and prompt before parsing. Prompt P1 asks, "Here is a question from a patient and two answers to it. Which answer do you prefer? Reply with one word only: A, B, or TIE." Prompt P2 adds clinical correctness, risk disclosure, and relevance as decision criteria used in CQA-Eval [9] and the restrictions proposed in prior work [1, 5]. For example, in P2, the prompt adds "do not reward a confident tone, ignore the order the answers appear in." Of 2,400 requested judgments, 2,396 are parseable. Luna under P1 and Sonnet under P2 each produce 398 valid judgments; the other four conditions produce 400 each. The preferences in Tab. 14 should not be interpreted as accuracy because the dataset does not include physician pairwise preferences as a reference.

Table 14: LLJ preference between the physician answer and a model answer from CQA-Eval under two prompts (%). *Model*: the judge preferred the model answer; *Physician*: it preferred the physician answer; *Tie*: it expressed no preference.

	Prompt P1			Prompt P2		
Judge Model	Model	Physician	Tie	Model	Physician	Tie
GPT-5.4 Nano	**43.0**	26.2	30.8	**54.2**	40.5	5.2
GPT-5.6 Luna	**58.0**	42.0	0.0	**53.5**	45.8	0.8
Claude Sonnet 5	**53.2**	44.5	2.2	44.7	**50.0**	5.3

We also find that LLJ preferences are sensitive to answer order, with preference reversal rates ranging from 11% to 40% when only the presentation order of the answers is swapped. Although P2 explicitly instructed judges to "ignore the order," Nano shows a higher reversal rate under P2 (40%) than under P1 (11%). In contrast, the other two LLJs showed lower reversal rates under P2, highlighting that explicit instructions to disregard answer order do not consistently mitigate positional bias across judge models.

B.2 Diagnosis Acceptance Rules

We examine evaluation-rule sensitivity using 160 `DeepSeek-V4-Flash` diagnosis predictions: 80 randomly sampled MedCaseReasoning cases and all 80 available MIMIC-CDM cases. Using ICD-10 coding assisted by Nano, we compare a strict rule requiring a match on normalized full codes with a broader rule matching the first three code characters. Switching from category-level matching to full-code matching reduces the measured accuracy by 18.8 percentage points, without changing any model prediction. These codes are not independently adjudicated by clinicians. The result therefore measures sensitivity to the acceptance rule, not the clinical superiority of either rule.

Tab. 15 clarifies how the three diagnosis examples in Tab. 2 should be interpreted under an explicit acceptance policy. These examples highlight both prompt sensitivity and inconsistencies with task-specific acceptance rules. Importantly, shared ICD-10 codes do not establish clinical equivalence, and benchmark acceptance does not imply universal interchangeability. Evaluation codebooks should explicitly define acceptable terminology variants, diagnostic specificity, etiologies, complication status, code granularity, and exclusion criteria.

Table 15: Interpretation of the diagnosis case studies under an explicit acceptance policy.

Diagnosis pair	Interpretation
Dx1: Urticaria Multiforme / Acute Urticaria	Subtype–broader diagnosis relation. Urticaria multiforme is a subtype of acute urticaria, but the diagnoses are not strictly equivalent. Acceptance depends on whether the task permits broader diagnoses. The LLJ judgment changes from incorrect to correct after adding a system prompt.
Dx2: Oculomotor Mononeuropathy / Post-Vaccinal Acute Oculomotor Nerve Palsy	Shared ICD-10 category (H49.0) does not establish equivalence. Post-vaccinal palsy specifies an etiology not supported by the broader diagnosis. The LLJ judgment changes from correct to incorrect after adding a system prompt.
Dx3: Cholecystitis / Perforated Gallbladder	Accepted under the MIMIC-CDM benchmark evaluation rule [12]. The LLJ rejects the pair both with and without the system prompt.

C Expert Effort and Evaluation Cost

The aim is not to replace scalable LLJ scoring with exhaustive clinician review. Instead, experts define a codebook, independently rate a stratified calibration sample, and adjudicate uncertain or high-severity cases. A validated judge can then evaluate the remaining outputs within its tested scope. Benchmark developers can make this process easier to reuse by releasing the codebook, prompt, judge version, inference settings, reference-sampling design, and validation results.

Let N be the number of outputs, k the number of expert ratings per output, m the calibration-set size, and e the proportion of the remaining outputs escalated to experts. Full human review uses Nk ratings. The hybrid design uses

$$mk + e(N - m)k \tag{1}$$

expert ratings, in addition to judge setup, robustness checks, inference, monitoring, and revalidation. If c_H denotes the cost per expert rating, those additional expenses must be less than the avoided review cost, $(N - m)(1 - e)kc_H$, for the hybrid approach to be less expensive overall. This is a cost-accounting condition, not empirical proof of savings.

Our structured sample of 51 studies also shows why a single cost estimate would be misleading. Among studies reporting positive human-panel sizes, the median is five raters (interquartile range 3–10, $n = 38$); the median clinician-panel size is also five (interquartile range 2–10, $n = 28$). For 31 studies whose human-rating unit is a generated output, the median rated sample is 400. For 10 studies whose unit is a case, question, or conversation, it is 97.5. Of the 42 studies that reported quantitative judge–human validation, 20 did not report human–human reliability.

Comparable monetary costs were rarely available: only one study reported an unambiguous human cost per item, and three reported an LLJ cost per item. In the single directly comparable example, PDSQI-9 clinical-summary evaluation took approximately 600 seconds and USD 50 in estimated physician labor, compared with 22 seconds and USD 0.05 in `GPT-o3-mini` API costs per summary. The comparison omits validation overhead and uses different types of cost, so it does not establish general cost-effectiveness. Future evaluations should report expert-hours, calibration coverage, adjudication and escalation rates, API costs, inference latency, and any monitoring or revalidation burden.

More capable judges, retrieval, or agentic workflows may reduce some current failure modes, but they do not eliminate the need for task-specific validation. Retrieved evidence may be outdated, irrelevant to a given patient, or incorrectly applied. Our proposal therefore retains the scalability benefit of LLJs while making their evidence and limitations visible before the scores inform consequential decisions.

D LLM Judges Prompts

D.1 Diagnostic Evaluation Prompt

The LLJ prompt used in Dx1-Dx3 in Tab. 2 is from MedCaseReasoning [23]. The optional system prompt we add in Tab. 2 is *"You are a professional medical expert that answers only 'y' or 'n'."*

MedCaseReasoning Prompt [23]

Is this predicted final diagnosis correct (y/n)?

Predicted final diagnosis: {predicted_diagnosis}
True final diagnosis: {actual_diagnosis}
Answer [y/n]

As introduced in Sec. 3.1, we further assess the different LLJ prompting strategy to investigate whether the prompt design affects diagnosis evaluation performance. As reported in Qiu et al. [13], they provided more detailed LLJ prompt with examples in the instruction.

Qiu et al. Prompt [13]

You are a professional medical diagnosis evaluation system. Now, you will receive two diagnosis results:

one is the diagnosis predicted by the model, and the other is the verified correct diagnosis . Your task is to judge whether the model-predicted diagnosis is correct.

When evaluating, please consider the following factors:

1.The same disease may have multiple aliases, for example, "Heart disease" may also be called "Cardiac disease".

2.There may be diversity in language expression, for example, "heart attack" and "myocardial infarction" may refer to the same disease.

3.Only judge whether the diagnosis result is correct, information such as the cause of the disease, symptoms, and treatment recommendations are not included in the evaluation scope.

4.If the correct diagnosis is included in the predicted diagnosis but some additional complications are mentioned, it is also considered correct.

Output Requirements

Only output your judgment result on the model-predicted [Diagnostic Decision Made by the LLMs] as "Correct|Wrong", do not output any other content.

Format to Follow:

[Correct|Wrong]

Below is the diagnosis result predicted by the model and the correct diagnosis:

[Predicted Diagnosis]

[Gold Diagnosis]

D.2 Free-form QA Evaluation Prompt

The LLJ prompt used in QA1-QA3 in Tab. 5 is from CQA-Eval [9].

CQA-Eval Prompt [9]

```
You are an expert medical evaluator. You will be presented with a medical question, an answer to that question, and an highlighted sentence from the answer.

You are scoring ONLY the information provided in the <mark>highlighted sentence</mark>. You may use the rest of the answer for additional context as necessary.

You will evaluate the quality of the highlighted sentence along the following 3 dimensions:

DIMENSIONS

1. The sentence aligns with current medical knowledge.
  To evaluate consider whether the answer:
  - Provides information based on existing scientific evidence
  - Expresses certainty in line with the current state of medical knowledge
  - Does not contain contradictions or logical flaws

2. The sentence addresses the specific medical question.
  - If the sentence directly addresses all the concerns included in the question -> assign **5**
  - If the sentence addresses some but not all concerns included in the question -> assign **4**
  - If the sentence provides necessary context or details to interpret the answer -> assign **4**
  - If the sentence  provides irrelevant details or context -> assign **2** or **1** depending on how irrelevant is the content

3. The sentence communicates contraindications or risks.
  To evaluate consider whether the answer:
  - Explains risks in clear, accessible language
  - Details potential consequences
  If there is NO mention of contraindications or risks in the highlighted sentence -> assign **Disagree**

You will express agreement with these 3 dimensions using a score from 1 to 5 where:
{
  5: "Agree"
  4: "Partially Agree",
  3: "Neutral",
  2: "Partially Disagree",
  1: "Disagree",
}

FOLLOW THESE STEPS TO MAKE YOUR DECISION:
- Read the medical question carefully
- Read the answer, focusing on the <mark>highlighted sentence</mark>
```

- Consider whether you agree or disagree with the given dimensions for the
↪ <mark>highlighted sentence</mark> and whether there are any minor inaccuracies or
↪ lack of sufficient nuance. Do not allow the length of the response to influence your
↪ evaluation. Be as objective as possible.
- Write your scores for <mark>highlighted sentence</mark> following the format of this
↪ example:
{
"The sentence aligns with current medical knowledge": 1,
"The sentence addresses the specific medical question": 3,
"The sentence communicates contraindications or risks": 5
}
The possible scores are: 5 for Agree, 4 for Partially Agree, 3 for Neutral, 2 for Partially
↪ Disagree, and 1 for Disagree. Do not use words to express agreement with the
↪ dimensions.

Please read the sentence carefully and score it thoughtfully. Use your best judgment! If you
↪ don't feel confident regarding your ability to score the sentence, feel free to use trusted
↪ reference materials such as clinical practice guidelines and systematic reviews.

D.3 Clinical Note Generation and Treatment Planning Evaluation Prompts

The LLJ prompts used in CNG1 and TP1 in Tab. 11 and Tab. 12 are from MedHELM [4].

Clinical Note Prompt [4]

You are a medical expert tasked with evaluating the quality of a generated clinical note based
↪ on a doctor-patient conversation. Your goal is to assess how well the note captures the
↪ clinical information from the conversation and compare it to the reference note (gold
↪ standard) in terms of accuracy, completeness and clarity.\\

The conversation will be provided in these tags:
<conversation>{{QUESTION}}</conversation>

The generated note will be provided in these tags:
<response>{{RESPONSE}}</response>

Carefully review the <response> based on the <conversation>.

For each of the following criteria, rate the response on a scale of 1 to 5 (1 = very poor, 5 =
↪ excellent), and provide a short justification for your score.

Evaluation Criteria:
Accuracy (1-5) - Does the note provide correct clinical information based on the conversation?
Completeness (1-5) - Does the note include all important medical details from the
↪ conversation?
Clarity (1-5) - Is the note written clearly and organized in a standard clinical format for
↪ clinicians

Output Format: Output the evaluation as a single valid JSON object matching the following
↪ structure:
{"accuracy":{"score":0,"explanation":"Explain why this score was given."},
"completeness":{"score":0,"explanation":"Explain why this score was given."},
"clarity":{"score":0,"explanation":"Explain why this score was given."}}

Ensure the output is valid JSON:
- Use double quotes (") for all keys and string values.
- When quoting text or sections inside the explanations, use escaped double quotes (") to maintain valid JSON formatting.
- Do not include any additional information in the output.

Treatment Plan Prompt [4]

You are a medical expert responsible for evaluating a proposed treatment plan based on provided patient information.

Your goal is to assess whether the treatment plan uses the patient's medical history, medications, symptoms, and other relevant details appropriately, follows clinical best practices, and aligns with the gold standard response in accuracy, completeness, and clarity.

The patient's information will be provided in these tags:
<patient_information>
{QUESTION}
</patient_information>

The proposed treatment plan will be provided in these tags:
<response>
{RESPONSE}
</response>

The gold standard treatment plan will be provided in these tags:
<gold_response>
{GOLD_RESPONSE}
</gold_response>

Carefully review the <response> based on the <patient_information> and compare it to the <gold_response> when needed.

For each category below, rate the response on a scale of 1 to 5 (1 = very poor, 5 = excellent) and provide a brief explanation for your score:

Evaluation Criteria:
Accuracy (1-5)
- Does the treatment plan provide correct medical advice based on clinical guidelines?

Completeness (1-5)
Does the treatment plan include all important medical details?

Clarity (1-5)
- Is the treatment plan written clearly so clinicians can easily understand it?

Output Format:
Generate a valid JSON object with the following structure:
```
{
  "accuracy": {
```

```
        "score": 0,
        "explanation": "Explain why this score was given."
    },
    "completeness": {
        "score": 0,
        "explanation": "Explain why this score was given."
    },
    "clarity": {
        "score": 0,
        "explanation": "Explain why this score was given."
    }
}

Ensure the output is valid JSON:
- Use **double quotes** (") for all keys and string values.
- When quoting text or sections inside the explanations, use escaped double quotes (\") to
↪  maintain valid JSON formatting.
- Do not include any additional information in the output.
```